

# CARISURG Triage Guardian Week 9 Safety

## Considerations One-Pager

Scope: Three HCI-specific and three HRI-specific safety considerations, each with a concrete failure mechanism and mitigation.

Triage Guardian is designed as an AI-assisted clinical decision-support system. The HCI pathway is the preferred nurse-facing implementation; the HRI pathway is an alternative pre-triage kiosk concept. The safety mechanisms below are specific to the interaction risks created by each setting.

| Setting | Safety consideration                 | Specific mechanism / risk                                                                                                                                                                                           | Design mitigation                                                                                                                                                                                                                              |
|---------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HCI     | Alarm fatigue                        | Repeated low-value or non-critical alerts during a busy or night shift can desensitise staff, increasing the chance that a genuinely critical alert is missed or acknowledged late.                                 | Use tiered alert severity; reserve the most visually prominent state for clinically meaningful high-risk changes; suppress duplicate alerts; require acknowledgement only for critical alerts; log unresolved alerts.                          |
| HCI     | Display legibility under stress      | Triage nurses may be interrupted, fatigued, or viewing the interface quickly. Small text, colour-only coding, crowded panels, or poor hierarchy can delay recognition of deterioration or lead to selection errors. | Use large readable labels for priority and waiting time; combine colour with text/icons; maintain strong contrast; place critical status near the patient identifier; avoid dense layouts and unnecessary scrolling.                           |
| HCI     | Automation bias / over-reliance      | Clinicians may accept the AI recommendation without independently reconciling it with the patient's clinical presentation, especially when workload is high.                                                        | Label outputs as decision support; show contributing factors and uncertainty; keep an obvious Accept/Modify/Override pathway; require clinician confirmation before final triage; retain override reasons in the audit trail.                  |
| HRI     | Proximity and physical accessibility | A physical kiosk can obstruct patient flow, be difficult to reach from a wheelchair, or create a collision/trip hazard in a crowded intake area.                                                                    | Use a fixed, stable footprint outside emergency egress routes; design wheelchair-accessible screen height and reach; avoid moving parts; preserve clear floor space around the unit; complete physical-site risk assessment before deployment. |

|     |                                         |                                                                                                                                                                                     |                                                                                                                                                                                                                                                 |
|-----|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HRI | Noisy-environment input failure         | If voice input is used, background noise, accents, masks, distress, or hearing impairment may cause incorrect symptom capture and therefore misleading downstream risk information. | Do not rely on voice as the sole input channel; provide touchscreen/text alternatives; display captured responses for patient confirmation; allow nurse correction; flag low-confidence or incomplete entries rather than guessing.             |
| HRI | Sensor failure and graceful degradation | Connected vital-sign sensors may fail, disconnect, or return implausible values. Treating those values as valid could trigger an incorrect risk alert or falsely reassure staff.    | Validate readings against plausible ranges; show sensor status; mark failed measurements as unavailable; never impute a normal value for a failed sensor; continue the intake workflow safely and route the patient to manual nurse assessment. |

### Cross-Cutting Safety Principle

The clinical authority still remains human. Neither the HCI dashboard nor the HRI kiosk should autonomously determine the final triage category. If AI support, network connectivity, or kiosk hardware becomes unavailable, the system must communicate the failure clearly and allow standard nurse-led triage to continue.